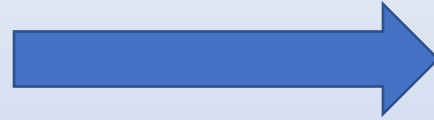


Introduction to Ceramics

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Ceramics:

Ceramics are compounds between **metallic and nonmetallic elements**

Most frequently: **oxides, nitrides, and carbides**

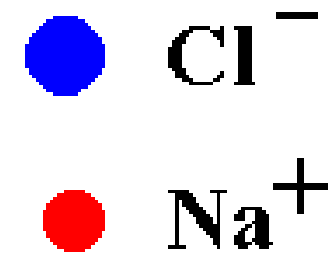
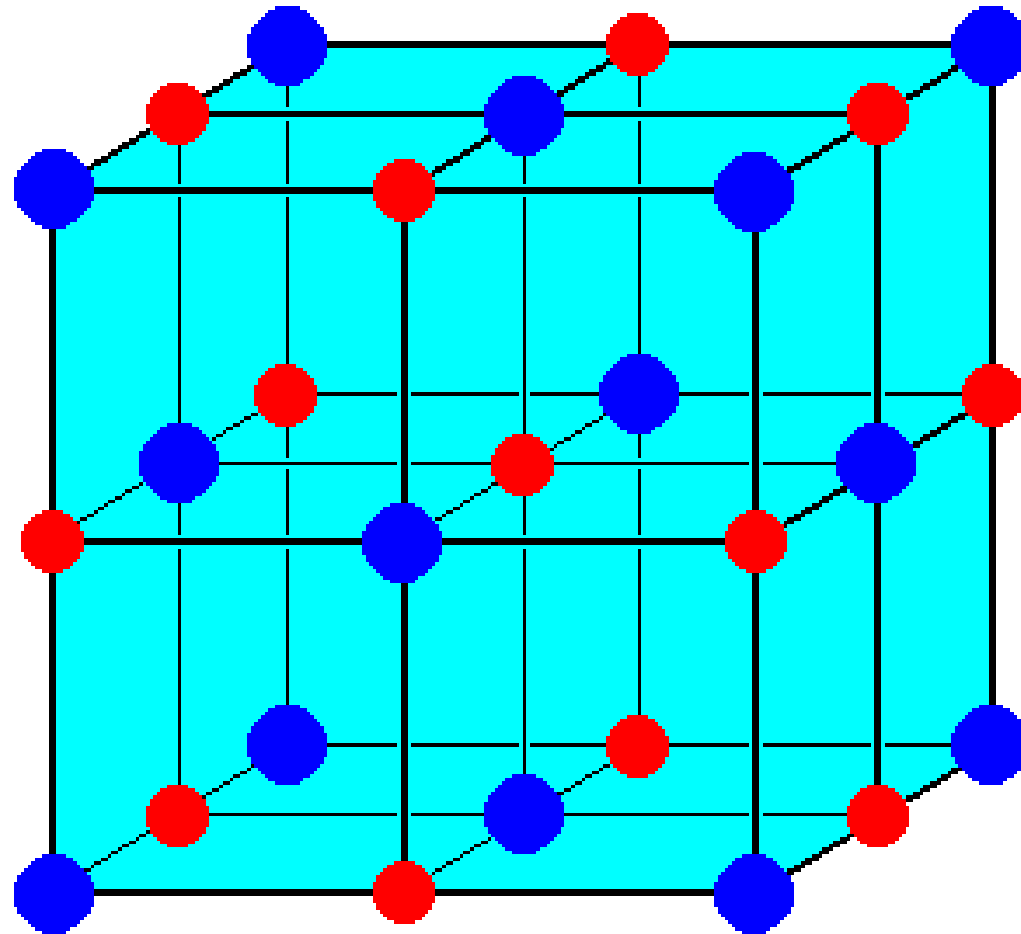
e.g.: **aluminum oxide (or *alumina*, Al_2O_3), silicon dioxide (or *silica*, SiO_2), silicon carbide (SiC), silicon nitride (Si_3N_4)**



- Ceramics can appear as either: **crystalline** or **amorphous solids**, the latter group being **called glasses**. 

In ceramics, **the negatively charged ions (*anions*)** are often different in size from the **positively charged ions (*cations*)**.

e.g. NaCl : chlorine ions are arranged as FCC unit cell crystalline structure and sodium ions occupy interstitial positions due to sodium ions small size



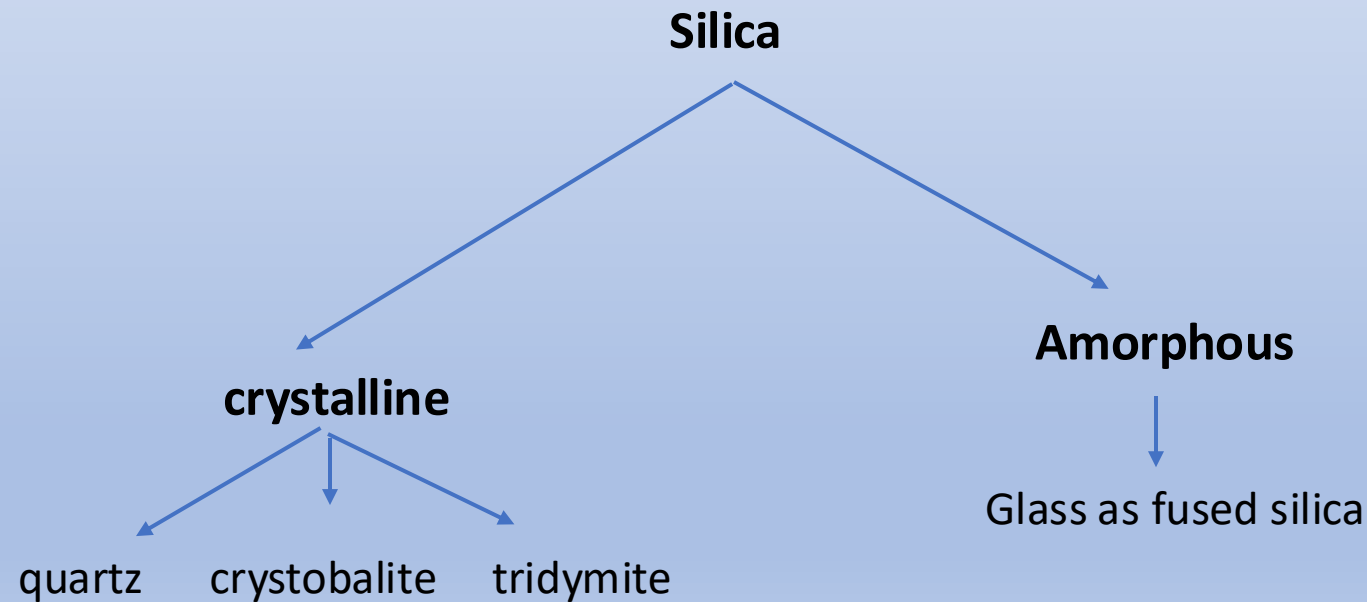
NaCl

NaCl unit cell (Source: <https://i.stack.imgur.com/HBKGW.gif>)



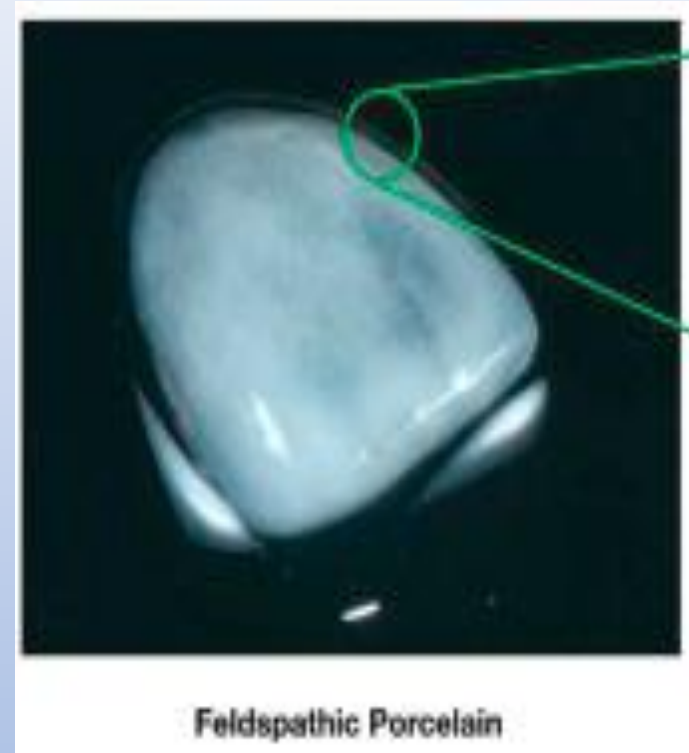
CERAMIC RAW MATERIALS:

Silica (SiO_2) forms the basis of many ceramics. Although it has a simple chemical formula, it can exist in many different forms.



Do you remember
polymorphism?

Aluminium oxide can be combined with silica form many complex ceramic formulations as **feldspathic glasses** which are used in ceramic restorations, and are compounds containing oxides of aluminium and silicon in combination with potassium, sodium or calcium (e.g. $\text{NaAlSi}_3\text{O}_8$).

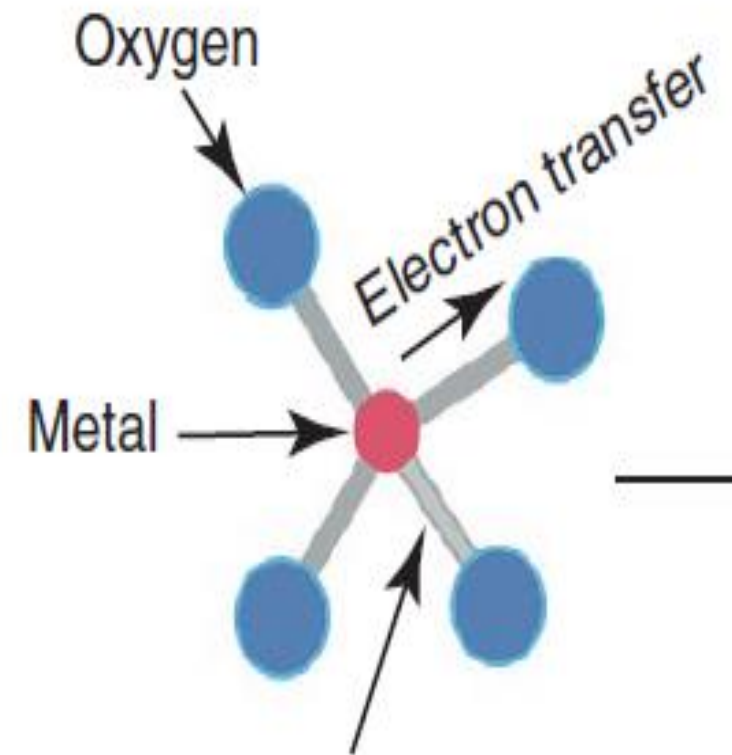


Chemical tempering of feldspathic porcelain for dentistry applications: A review

Author links open overlay panel, , , [Márcio Celso Fredel](#)^c , [Dachamir Hotza](#)^{c d}

Basic Ideas about Ceramics in Dentistry:

- At the atomic level, ceramics are composed of **metal–oxygen ionic bonds** (see Figure 14-1). Silicon (Si), zirconium (Zr), and aluminum (Al) are common metallic elements that occur in ceramics in combination with oxygen.
- Ionic bonds are **strong** and they do not tolerate bending. This intolerance  makes ceramics **brittle**.





- A second feature of ceramics is that the metal–oxygen ionic bonds occur in vast crystalline arrays (see [Figure 14-1B](#)).

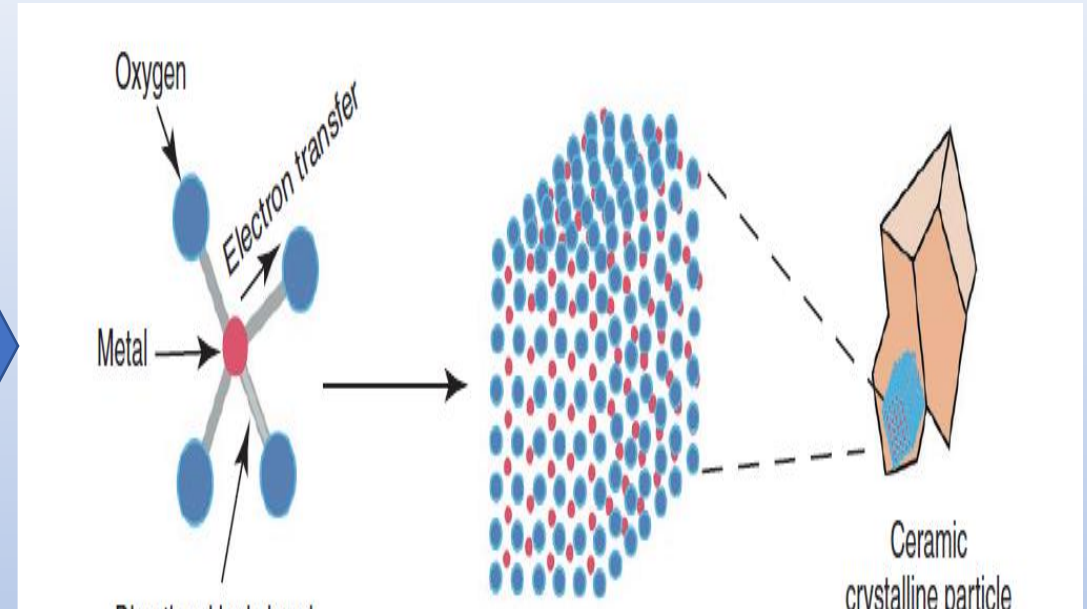


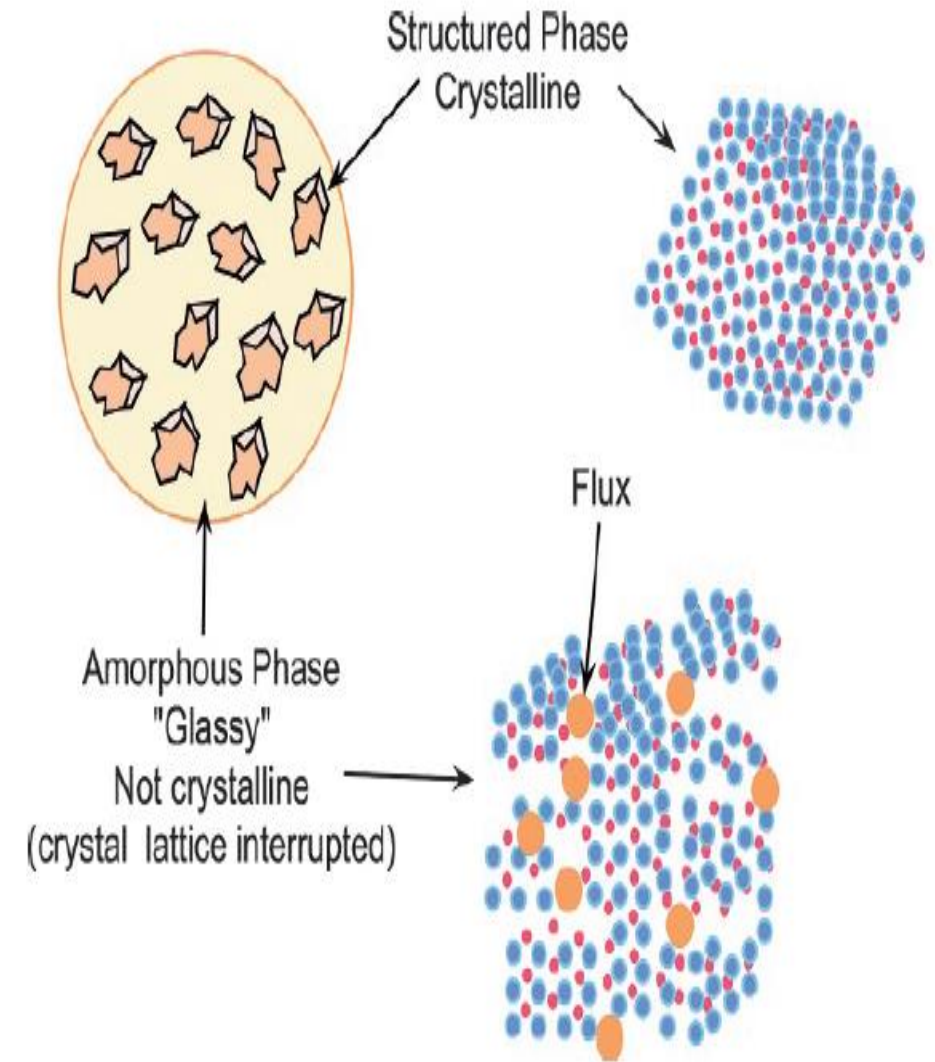
FIG 14-1B Ceramics are composed of arrays of metal atoms and oxygen held together by ionic bonds. These ionic bonds result from the transfer of electrons from the metal to oxygen, and they are strong (i.e., the bonds do not tolerate distortion). The metals and oxygen form arrays in a three-dimensional crystalline lattice that makes up the ceramic particle..



The continuity of the array may be **interrupted** by the presence of metal ions of sodium (Na) or potassium (K) for example.

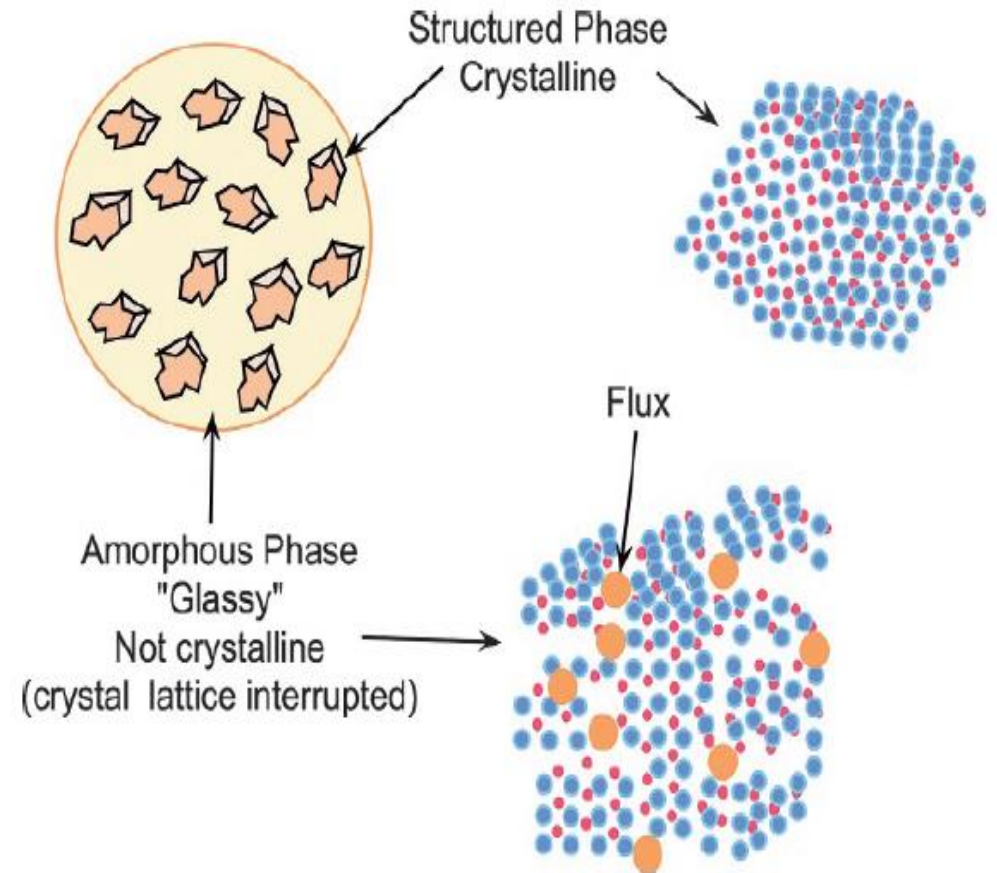
These interrupting ions are **called fluxes**.

Fluxes have several important effects on ceramic properties, **including reduced strength, lower fusing temperature, and more transparency**



When the crystalline array is disrupted sufficiently, it is described as **amorphous or glassy**

The **ratio between crystalline and amorphous areas** in a ceramic fundamentally determines how the ceramic will behave clinically in the mouth





The terms ceramic and porcelain are often used interchangeably, but **incorrectly**.

Ceramic refers to any material composed of the arrays of **metallic oxygen bonds described previously**.

on the other hand, Porcelain is a type of ceramic that results when **feldspar ($\text{K}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$) + silica (SiO_2) + and alumina (Al_2O_3)** are fired together with fluxes such as sodium carbonate (Na_2CO_3) or potassium carbonate (K_2CO_3)

During the firing of porcelain, large areas of amorphous ceramic are formed, **with small islands of a crystalline phase called leucite**.

Porcelains are often referred to as **feldspathic ceramics** in dentistry, and they are the **most esthetic** but **weakest** of the ceramics

**Porcelain = Feldspathic porcelain =
most esthetic = weakest**



<https://www.aegisdentalnetwork.com/cced/2011/04/feldspathic-veneers-what-are-their-indications>





Types and Uses of Ceramics in Dentistry Today (Figures 14-3 and 14-4).

Dental ceramics may be roughly classified into four types:

- 1-Traditional feldspathic (or glassy),
- 2- Glass dominated,
- 3-Crystalline dominated,
- 4-Crystalline

All are a consequence of the ratio of amorphous to crystalline phases and how the phases interact with one another.



1-Feldspathic: are porcelains that are composed primarily of: an **amorphous phase** (often called a **matrix**) + embedded **leucite crystals**

Leucite makes the **porcelain more opaque, stronger, and higher fusing and creates expansion upon heating**

Feldspathic ceramics also contain **small amounts of metal oxides** for **coloring** and **fluorescence** to optimize their esthetics.

Do you remember
fluorescing agents ?



Feldspathic ceramics remain among the most esthetic and weakest of the dental ceramics and are primarily used:

1- as veneers over alloys or ceramic substructures that impart strength to the restoration and support the weaker veneer



2- Feldspathic ceramics also are used for veneers that are bonded directly to tooth structure, particularly in the upper anterior region



Feldspathic ceramics can not be used as a full ceramic crown



2-Glass-dominated ceramics: contain increased amounts of crystalline phase relative to the glassy ceramics; crystals may be **leucite or fluoroapatite** (see Figures 14-3 and 14-4).

The increased crystalline phase gives the ceramic **higher strength** but sufficient translucency to serve in esthetic applications. ↓

The increased strength allows these ceramics to be used for:

- 1- anterior all-ceramic crowns that are not under excessive occlusal force .
- 2- They are excellent choices for veneering on alloys or ceramic cores
- 3- **laminate veneers** on anterior teeth

these materials cannot be used for posterior crowns or bridges.



https://www.researchgate.net/publication/362083179_Overview_of_Several_Typical_Ceramic_Materials_for_Restorative_Dentistry/figures?lo=1



https://www.researchgate.net/publication/26765039_Esthetic_rehabilitation_of_anterior_conoid_teeth_comprehensive_approach_for_improved_and_predictable_results/figures?lo=1

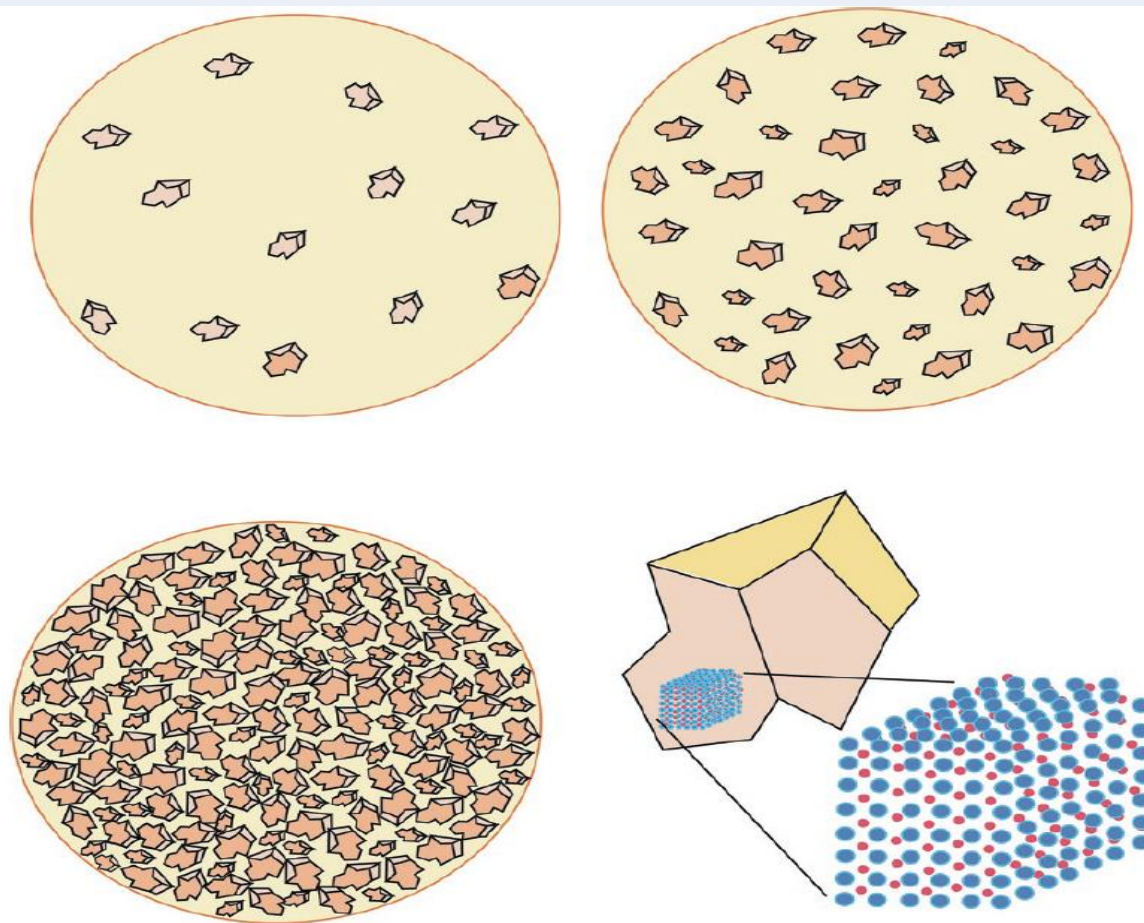


FIG 14-3 Ceramics in dentistry can be roughly divided into four categories. Feldspathic or glassy ceramics (*upper left*) are the oldest ceramic in dentistry. They consist of mostly amorphous glass with islands of a crystalline phase, often leucite. Glass-dominated ceramics (*upper right*) have mostly a glassy phase but have an increased abundance of crystalline phase that may be leucite or other compounds. Crystalline-dominated ceramics (*lower left*) have mostly a crystalline phase, which may be composed of several types of crystals but have a glassy phase that surrounds the crystals. Finally, crystalline ceramics (*lower right*) have no glassy phase; these ceramics are the newest and strongest of the ceramics used in dentistry.



3-Crystalline-dominated ceramics: are composed mostly (about 70 vol %) of a crystalline phase (see [Figure 14-3](#)).

The spaces between the crystals are occupied by an amorphous silica glass, and the two phases synergize to increase the strength of the ceramics.

The crystalline phase is generally: *spinel ($MgAl_2O_4$) or zirconia (ZrO_2), or alumina (Al_2O_3), or lithium disilicate*

Disadvantage: Crystalline dominated ceramics are much more opaque and therefore are far less esthetic.



Lithium disilicate ceramics: One exception for the previous disadvantage : is the lithium disilicate ceramics, its newer forms have been developed to provide sufficient esthetic properties for:

1- some all-ceramic posterior crowns

2- All ceramic anterior restorations (e.g. anterior bridge, anterior crowns, laminate veneers)



https://www.researchgate.net/publication/362083179_Overview_of_Several_Typical_Ceramic_Materials_for_Restorative_Dentistry/figures?lo=1



<https://www.thousandoaksfamilydentistry.com/blog/2014/5/28/emax-lithium-disilicate-crowns>



Crystalline dominated ceramics (spinel ($MgAl_2O_4$) or zirconia (ZrO_2), or alumina (Al_2O_3)), were used as cores for anterior or posterior all-ceramic crowns, onto which veneering ceramic (either glassy or glass-dominated) is added, But with caution as they could not bear very high occlusal forces at posterior teeth.



4-The crystalline ceramics are the newest and strongest of the ceramics used in dentistry (see [Figure 14-3](#)).

These ceramics are formed from either [alumina or zirconia](#) that has been seeded with other ions such as [magnesium or yttrium](#) to optimize their properties



Crystalline ceramics [have no glassy phase and are opaque](#).

They cannot serve as esthetic veneers on alloys.

[they are used as:](#)

- [1-high-strength cores, including posterior crowns and bridges,](#)
- [2- and for making full ceramic posterior crowns or bridges](#)
- [3- core for anterior crown or anterior bridges.](#)
- [4- recently translucent zirconia is used as full contour crowns for areas not subjected to high forces of mastication.](#)



Full-contour zirconia



A full-contour translucent zirconia crown



Porcelain fused-to-zirconia (PFZ)



<https://www.avfdentalgroup.com/blog/zirconia-crown-advantages-and-disadvantages-of-zirconia-crown/>

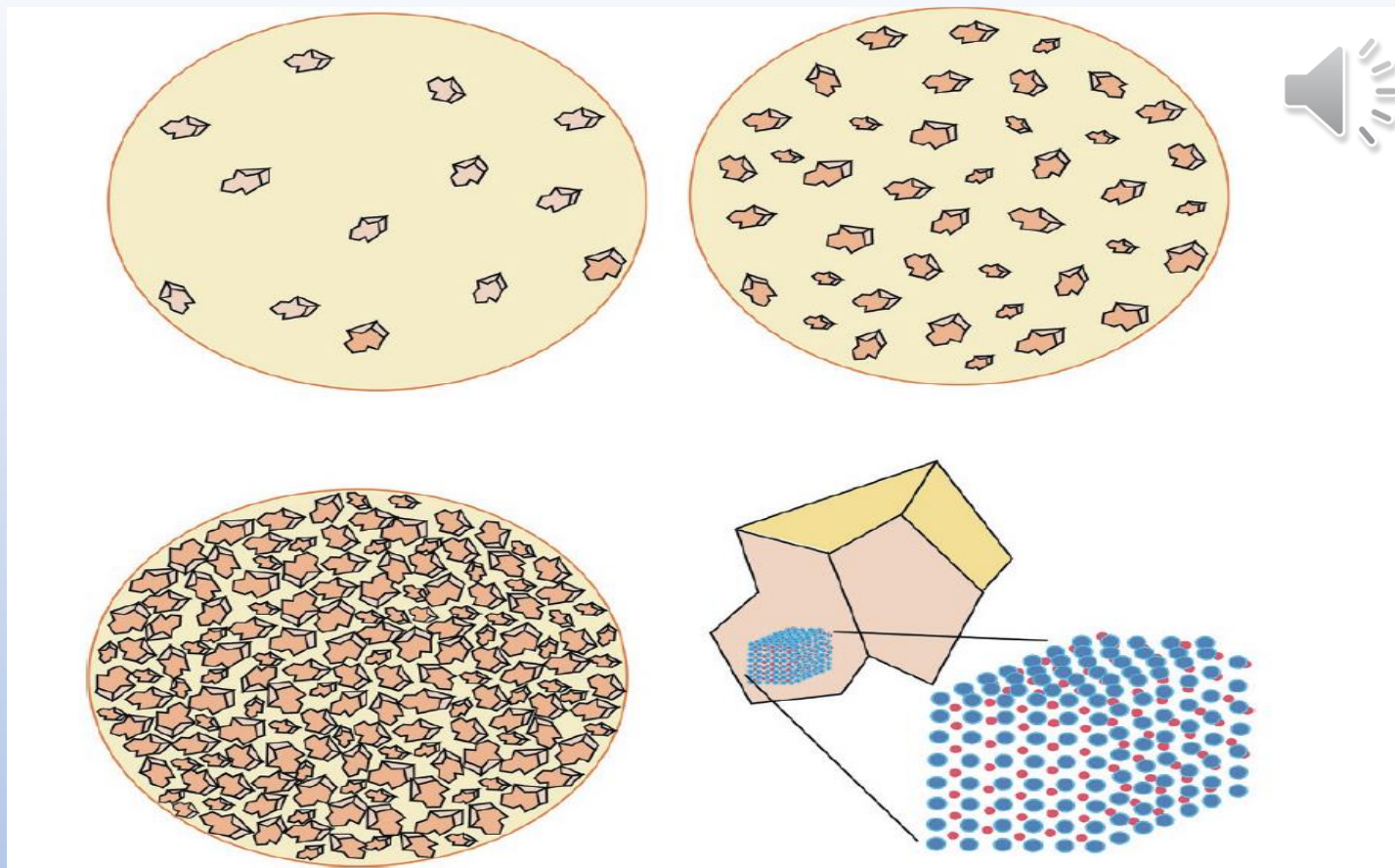
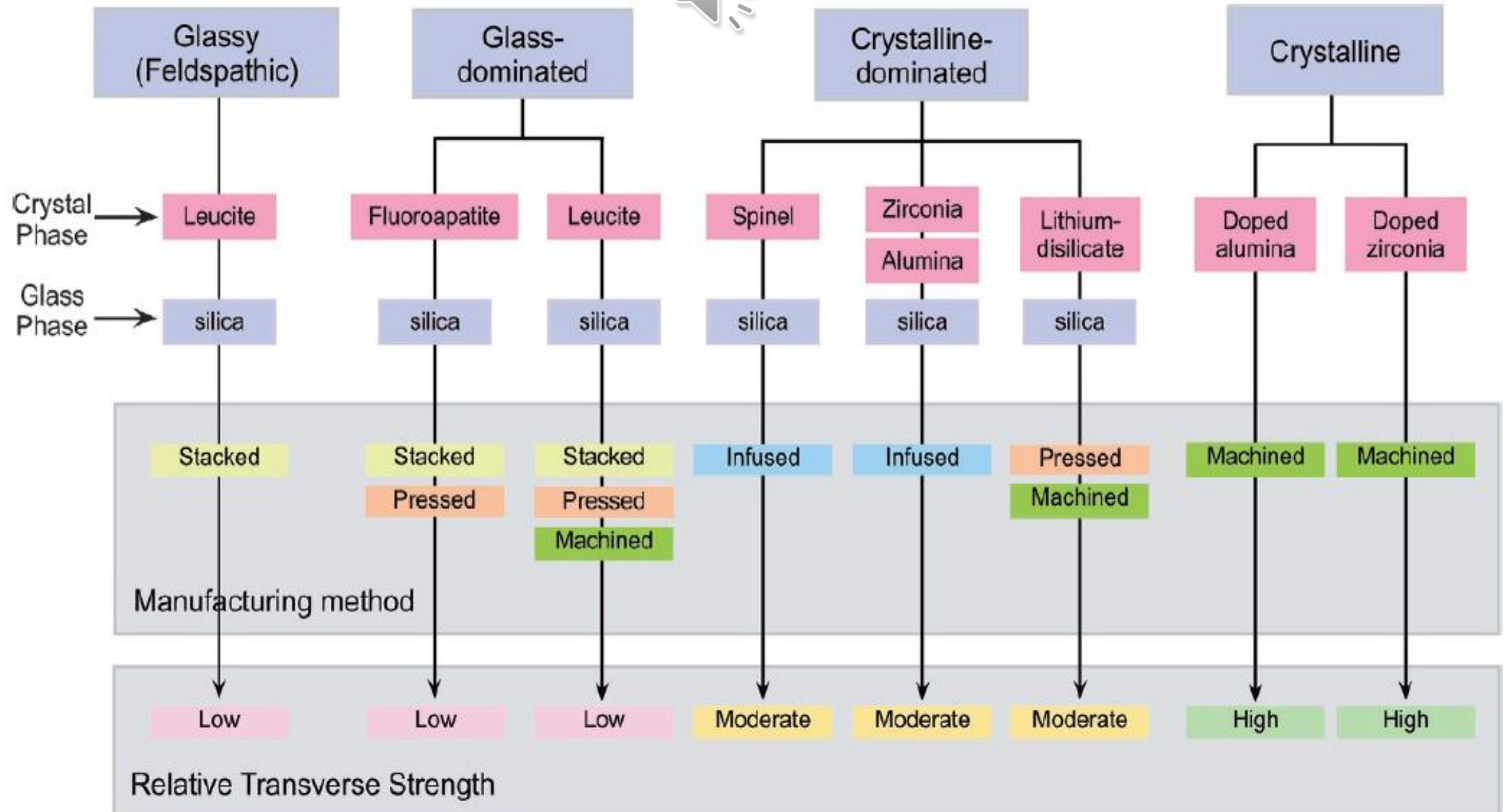


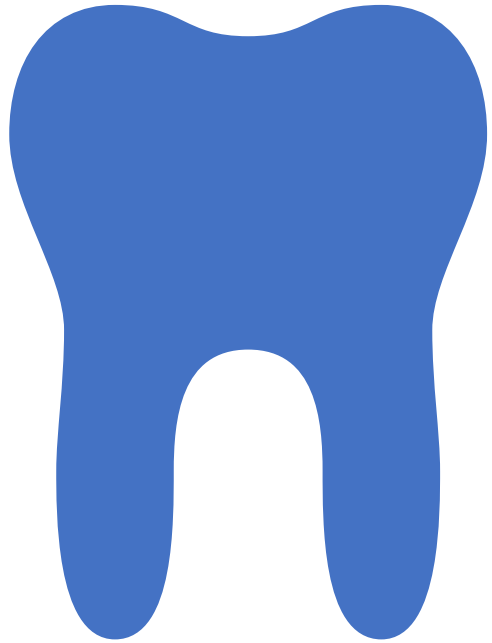
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Fabrication Techniques:

- One consequence of the strong ionic bonds in ceramics is that these materials melt only at relatively high temperatures.
- Some techniques are used to form the final dental restoration including stacking, infusing, pressing, or machining.







References

- References and photos from:
- 1- Dental materials foundations and applications 11th Edition
- 2- Craig's restorative dental materials 14th Edition.
- 3- pocketdentistry website